

Survival Analysis - Assignment 1

2026-02-17

Read packages

Q1

a)

Knowing that $h(t) = \lambda \gamma t^{\gamma-1}$, the cumulative hazard function $H(t)$ can be derived as

$$H(t) = \int_0^t h(u) du = \int_0^t \lambda \gamma u^{\gamma-1} du = \lambda u^\gamma \Big|_0^t = \lambda t^\gamma$$

The survival function $S(t)$ is correspondingly

$$S(t) = \exp(-H(t)) = \exp(-\lambda t^\gamma),$$

and the cumulative distribution function $F(t)$ is

$$F(t) = 1 - S(t) = 1 - \exp(-H(t)) = 1 - \exp(-\lambda t^\gamma).$$

b)

Given that $F(t) = p$ and $F(t) = \exp(-\lambda t^\gamma)$, we can solve the equation that,

$$\begin{aligned} 1 - \exp(-\lambda t^\gamma) &= p \implies \exp(-\lambda t^\gamma) = 1 - p \implies -\lambda t^\gamma \\ t^\gamma &= \ln(1 - p) \implies t^\gamma = \frac{1}{\lambda} \ln\left(\frac{1}{1 - p}\right) \implies t = \\ &= \left(\frac{1}{\lambda} \ln\left(\frac{1}{1 - p}\right)\right)^{\frac{1}{\gamma}} \end{aligned}$$

so the quantile function is $q(p) = \left(\frac{1}{\lambda} \ln \frac{1}{1-p}\right)^{\frac{1}{\gamma}}$.

Q2

a)

```
# raw data
times <- c(3,5,5,8,12,12,15,20,22,25)
status <- c(1,0,1,1,0,1,1,0,1,0)
df <- data.frame(times = times, status = status)

# build signature function
self_KM <- function(times, status){

  df_a <- aggregate(status ~ times, data = data.frame(times, status), FUN = sum)
```

```

colnames(df_a) <- c("Time", "n.event")

df_a$n.risk <- sapply(df_a$Time, function(t) sum(times >= t))

df_a$Survival <- cumprod(1 - df_a$n.event/df_a$n.risk)

# 95% CI
Var_logs <- cumsum(df_a$n.event / (df_a$n.risk * (df_a$n.risk - df_a
$n.event)))
log_es_surv <- log(df_a$Survival)
log_CI_lower <- log_es_surv - 1.96* sqrt(Var_logs)
log_CI_upper <- log_es_surv + 1.96* sqrt(Var_logs)
df_a$CI.lower <- exp(log_CI_lower)
df_a$CI.upper <- exp(log_CI_upper)

df_a <- df_a[df_a$n.event > 0, ]
return(df_a)
}

```

```

df_a <- self_KM(times, status)
print(df_a)

```

```

##   Time n.event n.risk  Survival   CI.lower  CI.upper
## 1     3         1     10 0.9000000 0.73200886 1.1065440
## 2     5         1      9 0.8000000 0.58681436 1.0906345
## 3     8         1      7 0.6857143 0.44471812 1.0573081
## 4    12         1      6 0.5714286 0.32583536 1.0021338
## 5    15         1      4 0.4285714 0.19308946 0.9512351
## 7    22         1      2 0.2142857 0.04331099 1.0602013

```

b)

```

fit_pack <- survfit(Surv(times, status) ~ 1, conf.type = "log")
summary(fit_pack)

## Call: survfit(formula = Surv(times, status) ~ 1, conf.type = "log")
##
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##    3     10         1   0.900  0.0949   0.7320    1.000
##    5      9         1   0.800  0.1265   0.5868    1.000
##    8      7         1   0.686  0.1515   0.4447    1.000
##   12      6         1   0.571  0.1638   0.3258    1.000
##   15      4         1   0.429  0.1743   0.1931    0.951
##   22      2         1   0.214  0.1748   0.0433    1.000

df_pack <- data.frame(Time = fit_pack$time,
                      Survival = fit_pack$surv,
                      CI.lower = fit_pack$lower,
                      CI.upper = fit_pack$upper)

```

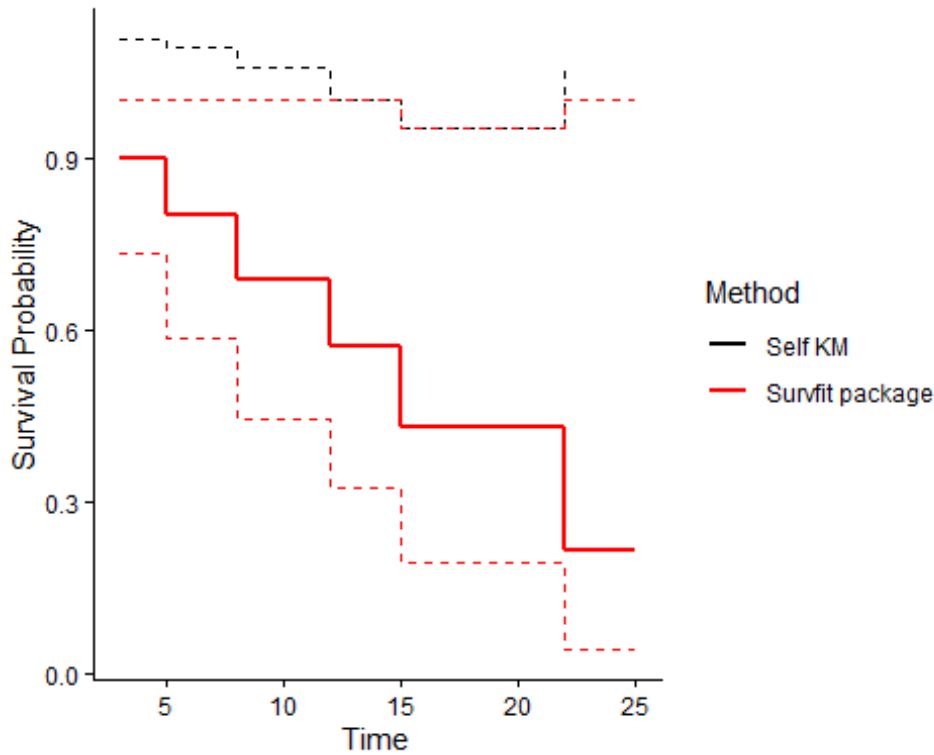
```

ggplot() +
  # self-KM
  geom_step(data = df_a,
            aes(Time, Survival, color = "A"),
            linewidth = 1) +
  geom_step(data = df_a,
            aes(Time, CI.lower, color = "A"),
            linetype = "dashed") +
  geom_step(data = df_a,
            aes(Time, CI.upper, color = "A"),
            linetype = "dashed") +

  # survfit package
  geom_step(data = df_pack,
            aes(Time, Survival, color = "B"),
            linewidth = 1) +
  geom_step(data = df_pack,
            aes(Time, CI.lower, color = "B"),
            linetype = "dashed") +
  geom_step(data = df_pack,
            aes(Time, CI.upper, color = "B"),
            linetype = "dashed") +

  scale_color_manual(
    name = "Method",
    values = c(A = "black", B = "red"),
    labels = c(A = "Self KM", B = "Survfit package")
  ) +
  labs(y = "Survival Probability") +
  theme_classic()

```



Q3

a)

The partial likelihood $L(\beta)$ can be expressed as:

$$L(\beta) = \prod_{i=1}^k \left\{ \frac{\exp(\beta^T x_i)}{\sum_{l \in R(t_i)} \exp(\beta^T x_l)} \right\}^{\delta_i}$$

$R(t_i)$ is “the risk set just before $t(i)$ ”. Intuitively, we first sort all observed times in ascending order. At a given time t_i , the risk set $R(t_i)$ consists of all individuals whose observed survival times are greater than or equal to t_i and have not experienced the event or been censored before t_i .

The likelihood for the i -th event is part of $L(\beta)$, which is

$$\left(\frac{\exp(\beta^T x_i)}{\sum_{l \in R(t_i)} \exp(\beta^T x_l)} \right)^{\delta_i}$$

and it can be explained as how likely a particular individual having the event at time t_i , relative to others in the risk set, given that one event occurs at time t_i .

b)

Given that $L(\beta) = \prod_{i=1}^k \left\{ \frac{\exp(\beta^T x_i)}{\sum_{l \in R(t_i)} \exp(\beta^T x_l)} \right\}^{\delta_i}$, we derive the score function following the steps below:

$$\begin{aligned} \log L(\beta) &= \sum_{i=1}^k \left(\beta^T x_i - \log \left(\sum_{l \in R(t_i)} \exp(\beta^T x_l) \right) \right) \\ S(\beta) &= \frac{d \log L(\beta)}{d \beta} = \sum_{i=1}^k \left(x_{(i),m} - \frac{\sum_{l \in R(t_i)} x_{(l),m} \exp(\beta^T x_l)}{\sum_{l \in R(t_i)} \exp(\beta^T x_l)} \right) \end{aligned}$$

Observing $S(\beta)$, we can find that, the second term is actually an expectation. The ratio $\frac{\exp(\beta^T x_l)}{\sum_{l \in R(t_i)} \exp(\beta^T x_l)}$ represents the weighted possibility that individual l having event at time t_i , given that one event occurs in $R(t_i)$. Then the second term can be simplified as $\sum_{l \in R(t_i)} x_{(l),m} \cdot p_l$, which is exactly the form of expectation $E(X_{(i),m})$. Setting $S(\beta)$ to zero, we obtain that, $\sum_{i=1}^k (X_{(i),m} - E(X_{(i),m})) = 0$. It implies that when the sum of actual covariates of individuals equals to the sum of expected value of covariates, the score functions equal to zero.

Q4

a)

The unscaled Schoenfeld residual for individual i and covariate j is that

$$r_{Sji} = \delta_i (x_{ji} - \hat{a}_{ji}) \cdot \frac{\sum_{l \in R(t_i)} x_{jl} \exp(\hat{\beta}^T x_l)}{\sum_{l \in R(t_i)} \exp(\hat{\beta}^T x_l)}$$

where δ_i is the event indicator (1 if event and 0 if censored), x_{ji} is the actual covariate value for individual i and $\hat{a}_{ji}$ is the risk-weighted average of covariate j among those who still at risk at t_i .

The scaled Schoenfeld residual is

$$r_{Si}^* = [d * \text{var}(\hat{\beta})] r_{Si}$$

where d is the total number of events in the dataset, $\text{var}(\hat{\beta})$ is the estimated variance-covariance matrix of the coefficients, and r_{Si} is the unscaled residual vector $x_{ji} - \hat{a}_{ji}$ for all covariates at event time t_i .

b)

i)

```
data_lung <- collett::lung
fit1 <- coxph(Surv(time, status) ~ factor(disease) + age,
```

```

      data = data_lung)
res_uns <- residuals(fit1, type = "schoenfeld")
#colnames(res_uns)
head(res_uns)

##      factor(disease)2 factor(disease)3 factor(disease)4      age
## 3          0.5042715        -0.2019231        -0.1101550    9.247792
## 4          0.5083320        -0.2035490        -0.1110419   12.322256
## 16         0.5158089        -0.2065429        -0.1126752   12.334432
## 21         0.5202029        -0.2083024        -0.1136351   -3.560494
## 22        -0.4760592         0.7902008        -0.1144516  -11.586077
## 25        -0.4784760         0.7942125        -0.1150326 -30.644897

mean(res_uns)

## [1] -5.370656e-14

res_sca <- residuals(fit1, type = "scaledsch")
head(res_sca)

##      factor(disease)2 factor(disease)3 factor(disease)4      age
## 3          3.2475949         2.4789086         0.6470900   0.09863604
## 4          3.4200728         3.5109231         0.8577665   0.14406554
## 16         3.4486421         3.4960389         0.8528683   0.14350098
## 21         2.6518213        -1.9040100        -0.2558313  -0.09390188
## 22         0.6388429         6.2905033         0.5085595   0.07394572
## 25        -0.3345421        -0.1223318        -0.8120944  -0.20899059

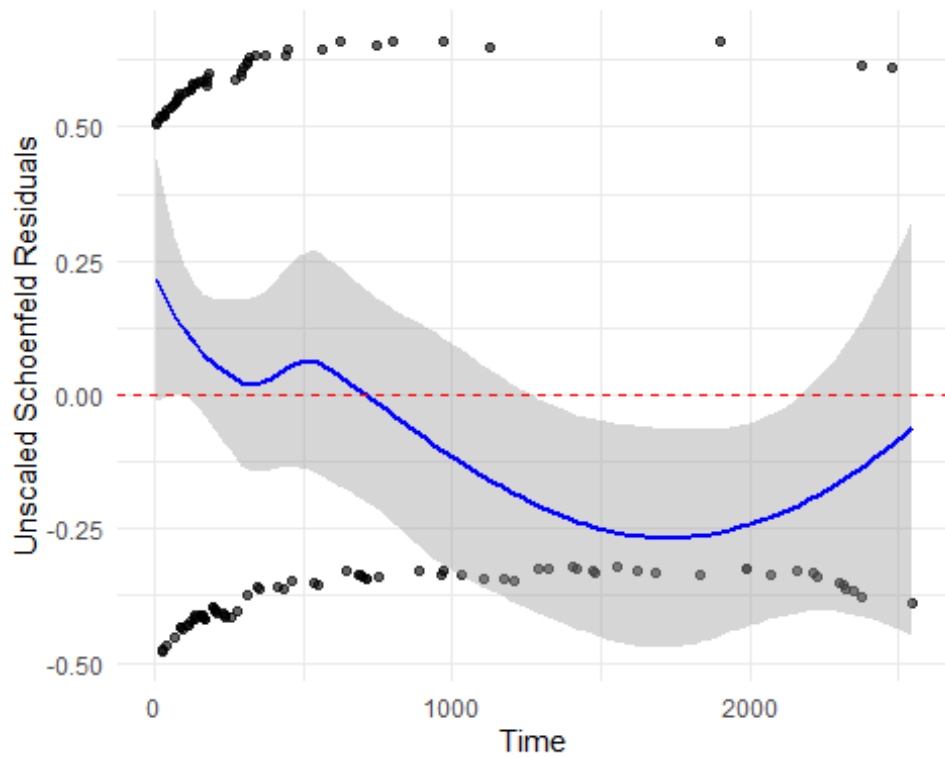
df_event <- data_lung[data_lung$status > 0, ]
event_time <- as.numeric(rownames(res_uns))
#head(collett::lung)
#length(res_uns[, 1])
res_df <- data.frame(
  time = event_time,
  res = res_uns[, 1]
)

res_df2 <- data.frame(
  time = event_time,
  res = res_sca[, 1]
)

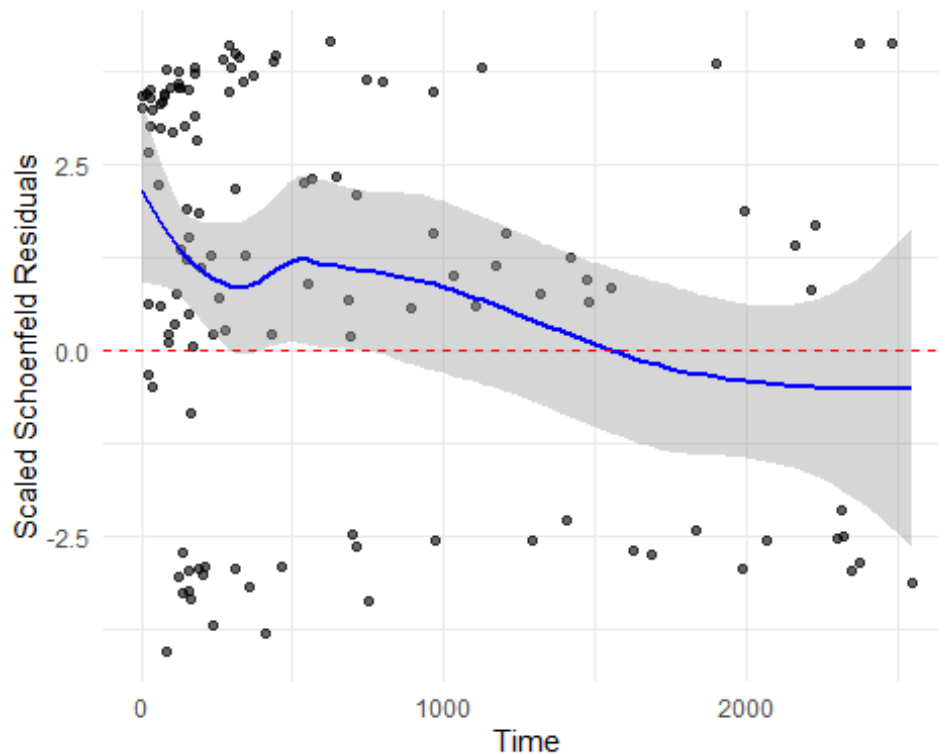
ggplot(res_df, aes(x = time, y = res))+
  geom_point(alpha = 0.6) +
  geom_smooth(method = "loess", se = TRUE, color = "blue") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")
+
  labs(x = "Time", y = "Unscaled Schoenfeld Residuals") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'

```



```
ggplot(res_df2, aes(x = time, y = res))+  
  geom_point(alpha = 0.6) +  
  geom_smooth(method = "loess", se = TRUE, color = "blue") +  
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")  
+  
  labs(x = "Time", y = "Scaled Schoenfeld Residuals") +  
  theme_minimal()  
  
## `geom_smooth()` using formula = 'y ~ x'
```



The unscaled Schoenfeld residual shows a noticeable bending trend, and the scaled Schoenfeld residual exhibits a decreasing trend, indicating that the proportional hazards assumption is not satisfied.

ii)

```
data_lung$distype1 <- as.numeric(data_lung$disease == 1)
data_lung$distype2 <- as.numeric(data_lung$disease == 2)
data_lung$distype3 <- as.numeric(data_lung$disease == 3)
data_lung$distype4 <- as.numeric(data_lung$disease == 4)

fit_tv <- coxph(Surv(time, status) ~ distype2 + distype3 + distype4 + a
  ge + tt(distype2),
               data = data_lung,
               tt = function(x, t, ...) x * log(t))
               #tt = function(x, t, ...) x * t)

summary(fit_tv)

## Call:
## coxph(formula = Surv(time, status) ~ distype2 + distype3 + distype4
+
##     age + tt(distype2), data = data_lung, tt = function(x, t,
##     ...) x * log(t))
##
## n= 196, number of events= 123
##
```

```
##               coef exp(coef) se(coef)      z Pr(>|z|)
## distype2      3.556425 35.037714  0.918486  3.872 0.000108 ***
## distype3      0.604414  1.830179  0.363108  1.665 0.096002 .
## distype4      0.414207  1.513171  0.317900  1.303 0.192593
## age           0.009063  1.009104  0.011061  0.819 0.412584
## tt(distype2) -0.474700  0.622071  0.158930 -2.987 0.002819 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## distype2      35.0377    0.02854    5.7905   212.0081
## distype3      1.8302     0.54639     0.8983     3.7288
## distype4      1.5132     0.66086     0.8115     2.8215
## age           1.0091     0.99098     0.9875     1.0312
## tt(distype2)   0.6221     1.60753     0.4556     0.8494
##
## Concordance= 0.636 (se = 0.028 )
## Likelihood ratio test= 25.72 on 5 df,   p=1e-04
## Wald test              = 23.29 on 5 df,   p=3e-04
## Score (logrank) test = 27.34 on 5 df,   p=5e-05

# vcov(fit_tv)
```

We are assuming that $\log_{HR}(t) = \beta_{distype2} + \beta_{time \cdot distype2} \cdot \log(t)$,
 $Var(\log_{HR}(t))$ is then

$$Var(\beta_d) + (\log(t))^2 Var(\beta_{d,t}) + 2\log(t) \cdot Cov(\beta_d, \beta_{d,t})$$

```
# extract coefficients and CI
idx_d  <- "distype2"
idx_tt <- "tt(distype2)"

b_d <- coef(fit_tv)[idx_d]
b_dt <- coef(fit_tv)[idx_tt]

var_d <- vcov(fit_tv)[idx_d, idx_d]
var_dt <- vcov(fit_tv)[idx_tt, idx_tt]
cov_dt <- vcov(fit_tv)[idx_d, idx_tt]

t_seq <- seq(1, 1500, length.out = 200)
log_t <- log(t_seq)

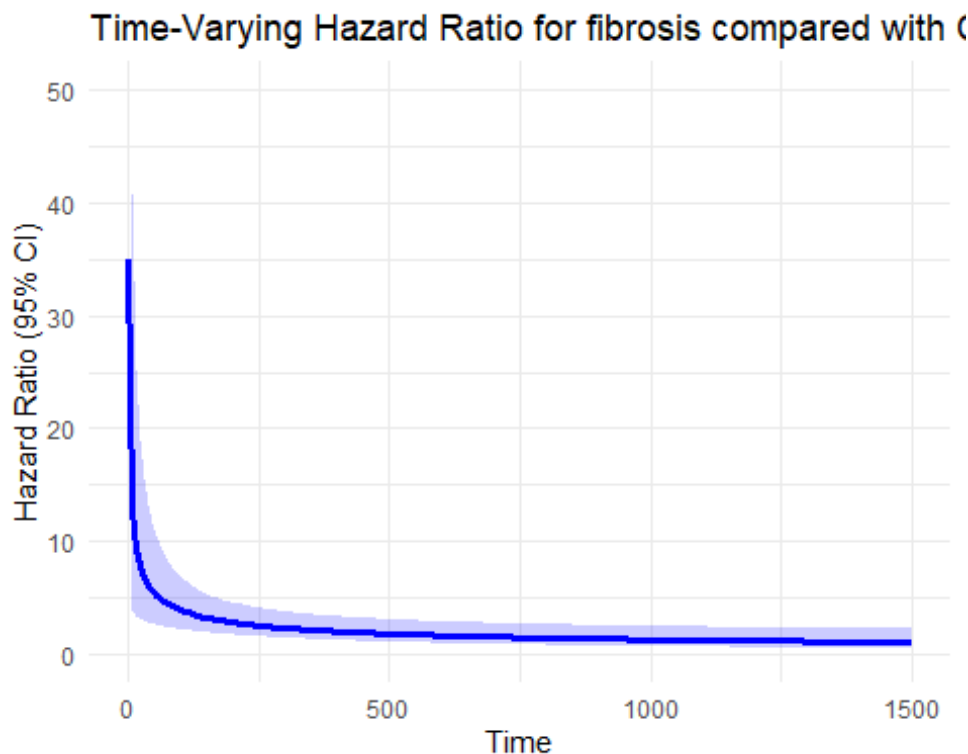
log_hr <- b_d + b_dt * log_t
se_log_hr <- sqrt(var_d + (log_t^2) * var_dt + 2 * log_t * cov_dt)

plot_df <- data.frame(
  time = t_seq,
```

```

HR = exp(log_hr),
lower = exp(log_hr - 1.96 * se_log_hr),
upper = exp(log_hr + 1.96 * se_log_hr)
)
ggplot(plot_df, aes(x = time, y = HR)) +
  geom_line(color = "blue", size = 1.2) +
  geom_ribbon(aes(ymin = lower, ymax = upper), fill = "blue", alpha = 0.2) +
  #geom_hline(yintercept = 1, linetype = "dashed", color = "#e41a1c", size = 0.8) +
  labs(
    title = "Time-Varying Hazard Ratio for fibrosis compared with COPD",
    x = "Time",
    y = "Hazard Ratio (95% CI)"
  ) +
  ylim(0,50)+
  theme_minimal()

```



Findings:

- 1) We can identify a significant time-varying effect by the significant p-value of `tt(distype2)` (0.002819), showing that the hazard ratio for fibrosis relative to COPD is not constant over time.
- 2) We can also see the trend of how HR changes over time: The HR is rather high initially, then it decreases rapidly and finally it converges to a low level.

Q5

a)

The likelihood for data tuple (u_i, v_i) in terms of probability is

$$L(i) = \prod_{i=1}^n \Pr(u_i < T_i \leq v_i \mid T_i > u_i) = \prod_{i=1}^n \frac{\Pr(u_i < T_i \leq v_i)}{\Pr(T_i > u_i)} = \prod_{i=1}^n \frac{F(v_i) - F(u_i)}{1 - F(u_i)}$$

Thus, log-likelihood is

$$l(i) = \sum_{i=1}^n \log \frac{F(v_i) - F(u_i)}{1 - F(u_i)}$$

Since $S(t) = 1 - F(t)$, the log-likelihood in terms of the survival function is

$$l(i) = \sum_{i=1}^n \log (S(u_i) - S(v_i)) - \log S(u_i)$$

Knowing that $h(t) = S(t) \cdot h(t)$, we obtain that

$$l(i) = \log \left(\int_{u_i}^{v_i} h(t) S(t) dt \right) - \log S(u_i)$$

Q6

a)

We know that

$$M(t) = N(t) - \int_0^t \lambda(u) du = N(s) - \int_0^s \lambda(u) du + N(t) - N(s) - \int_s^t \lambda(u) du \quad M(t) = M(s) + N(t) - N(s) - \int_s^t \lambda(u) du$$

Then $E(M(t) | \mathcal{F}_s)$ can be derived as below:

$$\begin{aligned} E(M(t) | \mathcal{F}_s) &= E \left(M(s) + N(t) - N(s) - \int_s^t \lambda(u) du \right) \\ &= M(s) + E \left(N(t) - N(s) - \int_s^t \lambda(u) du \right) \end{aligned}$$

Since $N(t) - N(s)$ follows Poisson distribution with parameter $\int_s^t \lambda(u) du$, the expectation of $E(N(t) - N(s))$ is the poisson parameter, which is

$$E\left(N(t) - N(s) - \int_s^t \lambda(u) du\right) = E\left(\int_s^t \lambda(u) du - \int_s^t \lambda(u) du\right) = 0$$

Thus, $E(M(t)|\mathcal{F}_s) = M(s)$ for all $s \leq t$, that is, $M(t)$ is a martingale.

b)

We know that $M(t) = M(s) + N(t) - N(s) - \int_s^t \lambda(u) du$, squaring this expression and setting $N(t) - N(s) - \int_s^t \lambda(u) du$ as ΔM , we get

$$M^2(t) = M^2(s) + \Delta M^2 + 2M(s)\Delta M$$

Given the conclusion in a) that $E(M(t)|\mathcal{F}_s) = M(s)$, taking the conditional expectation gives

$$\begin{aligned} E(M(t)^2 | \mathcal{F}_s) &= E(M^2(s) + (\Delta M)^2 + 2M(s)\Delta M | \mathcal{F}_s) \\ &= M^2(s) + E((\Delta M)^2 | \mathcal{F}_s) + 2M(s)E(\Delta M | \mathcal{F}_s) \end{aligned}$$

Then $E\left(M^2(t) - \int_0^t \lambda(u) du | \mathcal{F}_s\right)$ can be derived as following,

$$\begin{aligned} E\left(M(t)^2 - \int_0^t \lambda(u) du | \mathcal{F}_s\right) &= M^2(s) - \int_0^s \lambda(u) du + E\left((\Delta M)^2 + 2M(s)\Delta M - \int_s^t \lambda(u) du | \mathcal{F}_s\right) \\ &= M^2(s) - \int_0^s \lambda(u) du + E(\Delta M)^2 + 2M(s)E(\Delta M) - E\left(\int_s^t \lambda(u) du\right) \end{aligned}$$

Similar to question a), the expectation of $E(N(t) - N(s))$ is the poisson parameter,

$$E(\Delta M) = E(N(t) - N(s)) - \int_s^t \lambda(u) du = \int_s^t \lambda(u) du - \int_s^t \lambda(u) du = 0$$

Based on the variance equation that, $E(\Delta M)^2 = E\left(N(t) - N(s) - \int_s^t \lambda(u) du\right)^2 = \text{Var}(N(t) - N(s)) = \int_s^t \lambda(u) du$, we obtain that $E(\Delta M) - E\left(\int_s^t \lambda(u) du\right) = 0$, i.e.

$$E\left(M(t)^2 - \int_0^t \lambda(u) du | \mathcal{F}_s\right) = M^2(s) - \int_0^s \lambda(u) du$$

Consequently, we reach the conclusion that $M(t)^2 - \int_0^t \lambda(u) du$ is a martingale.

Q7

a)

First, the expression can be written as an integral:

$$|A(t) - A^*(t)| = \left| \int_0^t \alpha(s) ds - \int_0^t J(s) \alpha(s) ds \right| = \left| \int_0^t \alpha(s) (1 - J(s)) ds \right|$$

since $J(s) = I(Y(s) > 0)$ is an indicator function representing whether the risk set Y is empty at time s , $J(s) \in 0, 1$, $1 - J(s) \geq 0$. With the hazard function $\alpha(s) \geq 0$ too, the absolute value sign can be moved away, $|A(t) - A^*(t)| = \int_0^t \alpha(s) (1 - J(s)) ds$.

Next, we prove that $J(s) \xrightarrow{P} 1$. $Y(s)$ is the number of individuals at risk at time s , and it follows a binomial distribution, $Y(s) \sim \text{Bin}(n, p(s))$, where $p(s) = P(T \geq s)$ meaning the probability that an individual is still at risk at time s . Since $J(s)$ can only take the value 0 or 1, if $J(s) = 1$, $P(|J(s) - 1| \geq \epsilon) = P(0 \geq \epsilon) = 0$; if $J(s) = 0$, $P(|J(s) - 1| \geq \epsilon) = P(1 \geq \epsilon) = 1$. According to Law of Total Probability,

$$\begin{aligned} P(|J(s) - 1| \geq \epsilon) &= P(|J(s) - 1| \geq \epsilon | J(s) = 1) \cdot P(J(s) = 1) + P(|J(s) - 1| \geq \epsilon | J(s) = 0) \cdot P(J(s) = 0) \\ &= 0 + 1 \times P(J(s) = 0) \\ &= P(J(s) = 0) \end{aligned}$$

With the knowledge of binomial distribution, $P(J(s) = 0) = P(Y(s) = 0) = (1 - p(s))^n$. That is $P(|J(s) - 1| \geq \epsilon) = (1 - p(s))^n$, when n tend to inf,

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0 \quad \forall \epsilon > 0$$

i.e. $J(s) \xrightarrow{P} 1$.

As $J(s)$ is a non-increasing function, $J(s) \geq J(t)$ when $s \leq t$, that is, $1 - J(s) \leq 1 - J(t)$ when $s \leq t$.

$$\begin{aligned} |A(t) - A^*(t)| &= \int_0^t \alpha(s) (1 - J(s)) ds \\ &\leq \int_0^t \alpha(s) (1 - J(t)) ds \\ &= (1 - J(t)) \int_0^t \alpha(s) ds \\ &= (1 - J(t)) \cdot A(t) \end{aligned}$$

The corresponding probability is that,

$$\begin{aligned} P(|A(t) - A^*(t)| > \epsilon) &\leq P((1 - J(t)) \cdot A(t) > \epsilon) \\ &= P(J(t) = 0) \cdot P(A(t) > \epsilon) + P(J(t) = 1) \cdot P(0 > \epsilon) \\ &\leq P(J(t) = 0) = (1 - p(t))^n \end{aligned}$$

With $0 < 1 - p(t) < 1$, when n tend to inf.

$$\lim_{n \rightarrow \infty} P(|A(t) - A^*(t)| > \epsilon) = 0 \quad \forall \epsilon > 0$$

Finally, we got that

$$|A(t) - A^*(t)| \xrightarrow{P} 0.$$

Declaration of AI Usage: I used gemini to give me a structure of functions in R with the prompt “Show me the structure of a user-defined function in R” when building the “self-KM” function; I asked AI to debug for me in Q4 b ii) modeling for a time-varying model, it turns out factor(disease) cannot multiply log(t) directly, so I change disease into four dummy variables. No content, analysis, or interpretation was generated by the AI; the writing is entirely my own.